

Category.

Def. A category consists of

- A class of objects
- For each pairs of objects (A, B) a set $\text{Hom}_C(A, B)$ or simply $\text{Hom}(A, B)$ if C is clear) whose elements are called morphism. from A to B

- For each triple of object (A, B, C) we have a map

$$\begin{array}{ccc} \text{Hom}(A, B) \times \text{Hom}(B, C) & \longrightarrow & \text{Hom}(A, C) \\ (f, g) & \longmapsto & gf \end{array}$$

It is assumed that objects and morphisms satisfy the following conditions:

- If $(A, B) \neq (C, D)$ then $\text{Hom}(A, B) \cap \text{Hom}(C, D) = \emptyset$
- If $f \in \text{Hom}(A, B)$, $g \in \text{Hom}(B, C)$, $h \in \text{Hom}(C, D)$

$$(hg)f = h(gf)$$

- (Unit) For every object A we have an element $1_A \in \text{Hom}(A, A)$ s.t.
 $f 1_A = f$ for any $f \in \text{Hom}(A, B)$
 $1_A g = g$ for any $g \in \text{Hom}(B, A)$

Example: ① The category of sets objects. sets morphism. map.
denoted by Set

② The category of groups. objects groups morphism. group homo. Grp

③ The category of abelian groups object abelian groups.

morphism group homo. Ab .

④ R -module category objects. R -modules morphism. R -mod homo. $R\text{-Mod}$.

③ The category of f.g. modules $\text{obj: f.g. } R\text{-modules.}$
 $\text{mor: } R\text{-homo.}$ $R\text{-mod.}$

④ The category of topological spaces $\text{obj: topological spaces}$
 $\text{mor: constant map.}$ Top

Def: Subcategory A category $\mathcal{D}$ is called a subcategory of a category $\mathcal{C}$

if (i). the objects of $\mathcal{D}$ is a subclass of objects of $\mathcal{C}$

(ii). for any objects A, B of $\mathcal{D}$, we have

- $\text{Hom}_{\mathcal{D}}(A, B) \subseteq \text{Hom}_{\mathcal{C}}(A, B)$

- 1_A and the product of morphism for $\mathcal{D} \rightarrow$ the same as for $\mathcal{C}$

The subcategory $\mathcal{D}$ of $\mathcal{C}$ is called full if $\text{Hom}_{\mathcal{D}}(A, B) = \text{Hom}_{\mathcal{C}}(A, B)$
for any objects A, B of $\mathcal{D}$

Ex: Ab is full subcategory of Grp.

$R\text{-mod} \quad \dots \quad R\text{-Mod}$

Grp is not full of Set

Def. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. a covariant functor

$F:$ from $\mathcal{C}$ to $\mathcal{D}$, consists of

- for any object A of $\mathcal{C}$ there exist a unique object

$F(A)$ of $\mathcal{D}$

- for each pair (A, B) of objects of $\mathcal{C}$, F gives a map

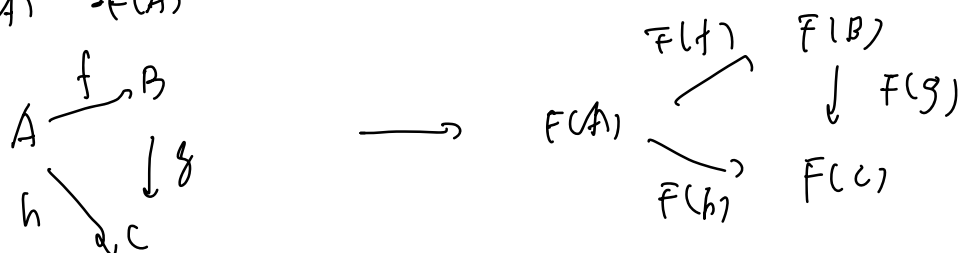
$$\text{Hom}_{\mathcal{C}}(A, B) \longrightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B))$$

$$f \longmapsto F(f)$$

It is assumed the following conditions are satisfied:

• If gf is defined in $\mathcal{C}$ then $F(g)F(f) = F(gf)$

• $F(1_A) = 1_{F(A)}$



M, N R -modules $\text{Hom}_R(M, N) = \{ f: M \rightarrow N \text{ is a } R\text{-hom} \}$
 $\downarrow$
 abelian group.

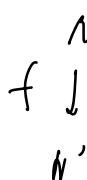
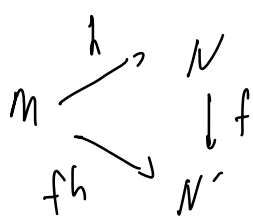
$$\text{End}_R(M) = \text{Hom}_R(M, M) \leadsto \text{ring}$$

E.g.: $M: R\text{-module}$ $\text{Hom}_R(M, -)$ is a functor

$$: R\text{-Mod} \rightarrow \text{Ab}$$

$$N \mapsto \text{Hom}_R(M, N)$$

$$f \mapsto \text{Hom}_R(M, f)$$



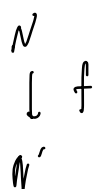
$$\begin{array}{ccc}
 \text{Hom}_R(M, N) & & \\
 \downarrow \text{Hom}_R(M, f): h \mapsto fh & & \\
 \text{Hom}_R(M, N') & &
 \end{array}$$

this is a covariant functor

E.g.: $M: R\text{-module}$

$$M \otimes_R - : R\text{-Mod} \rightarrow \text{Ab}.$$

$$N \mapsto M \otimes_R N$$



$$M \otimes N$$

$$\downarrow \text{id}_M \otimes f$$

$$M \otimes N'$$

$$M \otimes N$$

$$\downarrow$$

$$M \otimes f(N)$$

this is a covariant functor.

Def. Let $\mathcal{C}$ be a category Define $\mathcal{C}^{\text{op}}$ as.

$$\text{obj } \mathcal{C}^{\text{op}} = \text{obj } \mathcal{C} \quad \text{Hom}_{\mathcal{C}^{\text{op}}}(A, B) = \text{Hom}_{\mathcal{C}}(B, A)$$

if $f \in \text{Hom}_{\mathcal{C}^{\text{op}}}(A, B)$ $g \in \text{Hom}_{\mathcal{C}^{\text{op}}}(B, C)$ then gf in $\mathcal{C}^{\text{op}}$

$$= fg \text{ in } \mathcal{C}$$

we call $\mathcal{C}^{\text{op}}$ the dual category of $\mathcal{C}$

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow g & & \downarrow g \\ C & & C \end{array} \quad \begin{array}{ccc} A & \xleftarrow{f} & B \\ \uparrow fg & & \uparrow fg \\ C & & C \end{array}$$

in $\mathcal{C}^{\text{op}}$ in $\mathcal{C}$

Def. a contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ is a covariant functor from $\mathcal{C}^{\text{op}}$ to $\mathcal{D}$

Example: $\text{Hom}_R(-, M) : R\text{-Mod} \rightarrow \text{Ab}$

$$N \mapsto \text{Hom}_R(N, M)$$

$$\begin{array}{ccc} N & \xrightarrow{hf} & M \\ \downarrow f & & \downarrow h \\ N' & \xrightarrow{h} & M \end{array}$$

$$\begin{array}{ccc} N & & \\ \downarrow f & & \\ N' & & \end{array}$$

$$\begin{array}{ccc} \text{Hom}(N, M) & & \\ \uparrow \text{Hom}(f, M) : h \mapsto hf & & \\ \text{Hom}(N', M) & & \end{array}$$

Exactness ($R\text{-Mod}$)

Let M_i ($i \in \mathbb{Z}$) be R -modules

$\varphi_i : M_i \rightarrow M_{i+1}$ be R -homomorphism

We say $\dots \rightarrow M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} M_{i+1} \rightarrow \dots$ is a sequence of R -modules.

if $\text{Im } \varphi_{i-1} = \ker \varphi_i$, then we say the sequence is exact at M_i

A sequence $\dots \rightarrow M_{i-1} \rightarrow M_i \rightarrow M_{i+1} \rightarrow \dots$ is called exact if it is exact at any point.

A short exact sequence is $0 \rightarrow M' \xrightarrow{\tau} M \xrightarrow{\pi} M'' \rightarrow 0$

I is exact iff $\ker \tau = 0$ τ is mono

$$\operatorname{im} \pi = M'' \quad \pi \text{ is epi}$$

$$\operatorname{im} \tau = \ker \eta$$

$$\rightarrow M/M' \cong M''$$

Def R.S rings let $F: R\text{-Mod} \rightarrow S\text{-Mod}$ be a covariant functor

We say F is left exact if for any exact sequence

$$0 \rightarrow N \xrightarrow{\varphi} M \xrightarrow{\psi} P \quad \text{the sequence}$$

$$0 \rightarrow F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P) \quad \text{is exact.}$$

We say F is right exact if for any exact sequence

$$N \xrightarrow{\varphi} M \xrightarrow{\psi} P \rightarrow 0 \quad \text{the sequence}$$

$$F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P) \rightarrow 0 \quad \text{is exact}$$

Def: A contravariant functor $G: R\text{-Mod} \rightarrow S\text{-Mod}$ is called

left (or right) exact if it is left (or right) exact as

a covariant functor $G: (R\text{-Mod})^{\text{op}} \rightarrow S\text{-Mod}$

A functor is $\wedge$ exact if it is both left and right exact.

M is R -module

$\operatorname{Hom}_R(M, -): R\text{-Mod} \rightarrow \text{Ab}$ ($\mathbb{Z}\text{-Mod}$) is left exact

$\operatorname{Hom}_R(-, M): R\text{-Mod} \rightarrow \text{Ab}$ right exact.

$M \otimes_R -: R\text{-Mod} \rightarrow \text{Ab}$ right exact.

Def M is R -module

M is called a proj. inj if $\operatorname{Hom}_R(M, -)$ is exact. $\left. \begin{array}{l} \text{left} \\ \text{exact} \end{array} \right\}$ $\operatorname{Hom}_R(-, M)$ is exact

flat

$M \otimes_R -$

exact right mod

Since if R non-comm. $M \otimes_R -$ well defin need M right mod.

Bimodule

Def: Let R and S be rings. An abelian group M is called a left R -right S -bimodule in case M is both a left R -mod and right S -mod for which the two scalar multiplication jointly satisfy:

$$r(xs) = (rx)s \quad \forall r \in R, s \in S, x \in M$$

Notation ${}_R M$ left R -mod M_S right S -mod
 ${}_R M_S$ left R -right S -bimod.

Let R, S rings. $U = {}_R U_S$ bimodule

For each R -module ${}_R M$ there are two S -module.

$\text{Hom}_R({}_R U_S, {}_R M)$ left S -module

$\text{Hom}_R({}_R M, {}_R U_S)$ right S -module

$\forall s \in S \quad f \in \text{Hom}_R({}_R U_S, {}_R M)$

another: $f \in \text{Hom}_R({}_R M, {}_R U_S)$

$$s \cdot f : U \rightarrow M \\ m \mapsto f(us)$$

$$f \cdot s : M \rightarrow U \\ m \mapsto f(m)s$$

$\text{Hom}_R({}_R U_S, -) : R\text{-Mod} \rightarrow S\text{-Mod}$

$${}_R M \rightarrow \text{Hom}_R({}_R U_S, {}_R M)$$

Covariant

$f \mapsto ?$

$$\text{Hom}_R(-, R U_S) : R\text{-Mod} \rightarrow \text{Mod-}S \quad \text{contravariant}$$

$$RM \mapsto \text{Hom}_R(R U_S, RM)$$

$$\text{Hom}_R(R U_S, -) : R\text{-Mod} \rightarrow S\text{-Mod}$$

$$\begin{array}{ccccc} M & & \text{Hom}_R(R U_S, RM) & & h \\ \downarrow f & & \downarrow & & \downarrow \\ M' & & \text{Hom}_R(R U_S, RM') & & fh. \end{array}$$

Thm. The fun functor $\text{Hom}_R(R U_S, -) : R\text{-Mod} \rightarrow S\text{-Mod}$ is left exact

pf: let $0 \rightarrow N' \xrightarrow{f} N \xrightarrow{g} N''$ be exact in $R\text{-Mod}$
 Consider $0 \rightarrow \text{Hom}_R(R U_S, N') \xrightarrow{\text{Hom}_R(R U_S, f)} \text{Hom}_R(R U_S, N) \xrightarrow{\text{Hom}_R(R U_S, g)} \text{Hom}_R(R U_S, N'')$

• Show $\text{Hom}_R(U, f)$ is inj.

sp. $\varphi \in \text{Hom}_R(U, N')$ s.t. $\text{Hom}_R(U, f)(\varphi) = \neq \varphi = 0$.

if $u \in U$, $f(u) = 0$ then $\varphi(u) = 0$ since f inj

$\Rightarrow \varphi = 0 \Rightarrow \text{Hom}_R(U, f)$ is inj.

Next. show $\text{Im}(\text{Hom}_R(U, f)) \subseteq \ker(\text{Hom}_R(U, g))$

let $\varphi \in \text{Hom}_R(U, N')$ s.t. $\varphi \neq 0$ since $gf = 0$.

• Find we prove the reverse inclusion

let $\psi \in \ker(\text{Hom}_R(U, g))$ i.e. $\psi \in \text{Hom}_R(U, N)$ and

$g\psi = 0$.

$\forall u \in U \quad g\psi(u) = 0 \rightsquigarrow \psi(u) \in \ker g = \text{Im } f$

$$\exists y \in N \text{ s.t. } f(y) = \varphi(x)$$

f inj $\Rightarrow y$ is unique determined by $\varphi(x)$

Define a map $U \rightarrow N'$, $x \mapsto y$

$$\text{check } f \circ \varphi = \varphi \Rightarrow f = \text{Hom}(U, f)$$

Thm. The contravariant hom functor $\text{Hom}_K(-, K U_S) : R\text{-Mod} \rightarrow \text{Mod-}S$ is left exact.

Bimodule ${}_S M_K$, ${}_R N$, $M \otimes_K N$

For each $s \in S$ the mapping $M \times N \rightarrow M \otimes_K N$ balanced
 $(m, n) \mapsto (sm) \otimes n$

so there is a unique bilinear map homo

$$v(s) : M \otimes_K N \rightarrow M \otimes_K N$$

$$m \otimes n \mapsto (sm) \otimes n$$

check. $v : S \rightarrow \text{End}(M \otimes_K N)$ is a ring homo.

m_R . ${}_R N_T$: $M \otimes_K N$ is a right T -module.

$$(m \otimes n) t = m \otimes (nt)$$

Lem. If ${}_S M_R$ and ${}_R N_T$ are bimodules, the $M \otimes_K N$ is a left S -right T -bimodule with.

$$s(m \otimes n) = (sm) \otimes n \quad (m \otimes n) t = m \otimes (nt)$$

Let m_R , m'_R , ${}_R N$, ${}_R N'$

let $f : M \rightarrow M'$, $g : N \rightarrow N'$ be R -homo.

$$M \times N \rightarrow M' \otimes_R N' \text{ balanced}$$

$$(m, n) \mapsto f(m) \otimes g(n)$$

so $f \otimes_R g : M \otimes_R N \rightarrow M' \otimes_R N'$

$(f \otimes g) (m \otimes n) \mapsto f(m) \otimes g(n)$ is a grp homo.

Lem: $M_R, M'_R, R^N, R^{N'} \quad f_1, f_2, f \in \text{Hom}_R(M, M')$
 $g_1, g_2, g \in \text{Hom}_R(N, N')$

Then (1). $(f_1 + f_2) \otimes g = f_1 \otimes g + f_2 \otimes g$

(2). $f \otimes (g_1 + g_2) = f \otimes g_1 + f \otimes g_2$

(3). $f \otimes 0 = 0 \otimes g = 0$

(4). $1_M \otimes 1_N = 1_{M \otimes N}$

Lem. $M_R, M'_R, M''_R \quad R^N, R^{N'}, R^{N''}$

$f: M \rightarrow M', f': M' \rightarrow M'', g: N \rightarrow N', g': N' \rightarrow N''$

Then: $(f' \otimes g') (f \otimes g) = f' f \otimes g' g$

Let $U = {}_S U_R$ bimodule

Def. $U \otimes_R - : R\text{-Mod} \rightarrow S\text{-Mod}$ functor

$M \mapsto U \otimes_R M$

$f \mapsto \text{id}_U \otimes f$

Similarly define $- \otimes_R U$ for $R U_S$

$\text{Mod-}R \rightarrow \text{Mod-}S$

$N \mapsto N \otimes_R U$

$g \mapsto g \otimes \text{id}_U$

prop (1). ${}_S U_R$ Then $U \otimes_R - : R\text{-Mod} \rightarrow S\text{-Mod}$ is a covariant functor

(2). $R U_S$ Then $- \otimes_R U : \text{Mod-}R \rightarrow \text{Mod-}S$ is a covariant functor

Thm: s.t.p. the tensor functor $u \otimes_R - : R\text{-Mod} \rightarrow S\text{-Mod}$ is right exact.

$$N' \xrightarrow{f}, N \xrightarrow{g}, N'' \longrightarrow 0 \quad \text{exact}$$

$$\Rightarrow u \otimes_R N' \xrightarrow{\text{id} \otimes f}, u \otimes_R N \xrightarrow{\text{id} \otimes g}, u \otimes_R N'' \longrightarrow 0 \quad \text{exact.}$$

$$\text{if } g \text{ epi} \Rightarrow \text{id} \otimes g \text{ epi}$$

$$\forall x \in u \otimes_R N \quad x = \sum_i u_i \otimes n_i$$

$$(\text{id}_u \otimes g)(x) = \sum_i u_i \otimes g(n_i)$$